ExoInstruments

A short guide to your first exoplanet surveys in Kerbal Space Program

ExoInstruments turns KSP into an observational astronomy program. You point real telescopes at real stars, collect simulated data with realistic noise, and dig planet signals out of that data the same way professional astronomers do. This guide walks you through your first surveys without assuming you know any of the science. Each concept gets a short explanation the first time it appears.

What you need installed

The mod itself, plus ModuleManager and Kerbal Konstructs. If you installed through CKAN, both came with it. BetterTimeWarpContinued is optional but very useful, since surveys cross weeks of in game nights and faster time warp makes that painless.

1. Opening the observatory

From the space center, click the ExoInstruments button in the toolbar. The observatory opens as a fullscreen panel. Everything in the mod happens here. The left column is where you pick a target and an instrument. The right column shows the sky, your data, and the analysis results.

2. Picking a target star

The sky chart shows the real night sky, drawn from a real star catalog. Every point is an actual star with its true position, brightness, and color. Click a star to select it, or type part of a name in the filter box to highlight matches on the chart.

Two things matter most when choosing your first target:

- **Brightness.** Star brightness is measured in magnitude, and the scale runs backwards: lower numbers mean brighter stars. A magnitude 8 star is bright and easy, a magnitude 13 star is faint and hard. Brighter targets give cleaner data, because more light reaching the telescope means less relative noise. For a first survey, stay below magnitude 11 or so.
- **What the catalog says.** Outside career mode the target card tells you whether the star has a known planet and how it was found. A planet discovered by the transit method is a planet you can also find with the transit method here. In career mode this information is hidden until you survey the star yourself, which is the whole point of the survey.

3. Choosing an instrument

Below the target card sits the observatory selector. Each entry is a real facility with its real performance numbers, and clicking the About button under the selector shows a short presentation of the device. The instruments split into three families, one per detection method:

- **Transit photometry** (SPECULOOS, WASP, TESS). The telescope watches a star's brightness over time. If a planet's orbit happens to carry it across the face of its star as seen from here, the star dims by a tiny fraction, then recovers. That periodic dip is the signal.

- **Radial velocity** (SOPHIE, HARPS, ESPRESSO). A spectrograph measures the star's motion toward and away from us. An orbiting planet tugs its star in a small circle, and that wobble shows up as a periodic shift in the star's spectrum, measured in meters per second.
- **Direct imaging** (ELT). The telescope tries to photograph the planet itself, a faint dot beside a star millions of times brighter. Only the largest telescope in the roster can attempt this at all.

Start with SPECULOOS. It is unlocked from the beginning, it is the most precise transit instrument in the roster, and transits are the most visual way to learn the loop.

4. Your first transit survey, step by step

Pick a bright star that the catalog lists with a transiting planet (outside career mode the feasibility hints under the instrument selector will tell you what to expect). Then:

1. Check the observing forecast under the Start Observation button. It is a color grid: each row is one upcoming night, each column a time of night. Red means the telescope cannot usefully observe then (daytime, target below the horizon, or a moon in the way). Blue means prime conditions. Click any good cell and the game warps to that moment.
2. Press **Start Observation**. The telescope now takes an exposure every 30 seconds of game time, whenever the sky allows it. Ground based telescopes only collect at night with the target above the horizon, so your data will have gaps. That is normal, real light curves have the same gaps.
3. Engage time warp. Data accumulates as game time passes. The plot on the right fills in as points arrive. A few in game days of coverage is a good first baseline, and more nights always help.
4. Press **Stop Observation**, then **Run Detection Analysis**. The detector searches the whole light curve for periodic dips. On a large dataset this takes a while, sometimes minutes. The label under the button shows elapsed time so you know it is working. The game stays responsive during the search.
5. Read the scan report. If a planet was found you get a phase folded plot: every measurement rearranged onto a single orbit so all the transits stack on top of each other. A clear dip in that plot is your planet.

How to read the numbers

Period is how long the planet takes to orbit, in days. **Depth** is how much the star dims during transit, in parts per million; a depth of 10000 ppm means the star loses one percent of its light. Deeper means a bigger planet relative to its star. **SNR** is the signal to noise ratio, the confidence of the detection. The detector requires SNR 8 to claim anything. An SNR near 8 is marginal, 20 or more is solid. The report also converts depth into an implied planet radius, so you can see what kind of world you caught.

> If the analysis reports nothing, that is a real result too. Maybe the baseline was too short to cover two or three orbits, maybe the star is too faint for the instrument, or maybe there is genuinely nothing transiting. Longer baselines and brighter targets are the two levers that matter.

5. Your first radial velocity survey

Radial velocity work feels different: instead of thousands of brightness points per night you collect one velocity measurement every several hours, and the campaign runs for weeks. The mod tells you up front how long. When you select an RV instrument on a known system, it estimates the baseline needed to resolve the longest orbital period in the system, and during the session a button offers to warp you to that suggested baseline directly.

After the analysis, the report lists each detected signal with its period and its velocity amplitude K in meters per second, plus the implied minimum planet mass. Multi planet systems come out one signal at a time: the strongest is found and subtracted, then the residuals are searched again, which is exactly how real RV surveys separate planets.

> One warning worth knowing: the star itself is noisy. Spots and magnetic activity make every star jitter by a meter per second or more, and no instrument upgrade removes that. If a target card shows high stellar activity, small planets around it are genuinely harder to confirm. That is not a bug, it is the actual reason finding a second Earth is difficult.

6. Career mode: the survey program

In a career save the mod plays as a discovery loop. Star identities are hidden until you survey them: an unscanned target is just a position and a brightness, exactly what a real telescope sees. The catalog is also salted with ordinary background stars that host no known planet, so a null result is always possible and finding something always means something.

Economics: every observation costs Funds as telescope time, and each detection pays Science. Bigger instruments cost more per pointing but pay a larger reward multiplier. New instruments unlock with Funds plus a track record of Science earned from your own scans, so the path forward is: survey cheap targets with SPECULOOS, build up Science, then buy your way up the roster. Locked instruments are listed in the observatory selector with exactly what they still require.

7. Where to go after the basics

- **Multi planet transits.** Compact systems stack several transit signals in one light curve. The analysis peels them apart automatically and reports each one as a separate signal.
- **Transit timing variations.** With a long enough baseline, the report checks whether transits arrive slightly early or late in a repeating pattern. That pattern is the gravitational fingerprint of another planet pulling on the one you see, and it earns a bonus Science award.
- **The Rossiter McLaughlin effect.** Run an RV campaign on a star whose transiting planet you already identified, and the spectrograph automatically samples each transit window at high cadence. The analysis then measures whether the planet orbits in the same plane the star spins in, which is real evidence about how the system formed.
- **Direct imaging with the ELT.** The endgame instrument. The panel predicts how many hours of integration a 5 sigma detection needs before you commit, and even a failed companion search still characterizes the star itself.

Good hunting. Every system in the catalog is real, so anything you find in the mod exists, somewhere out there, for real.